

PREDICTABILITY OF VIRTUAL (PREDICTION) MARKETS

INTRODUCTION:

Prediction markets are platforms where people trade contracts based on future events. Prices reflect collective belief and are interpreted as probabilities. Participants buy and sell contracts tied to outcomes. Prices move based on new information and trader beliefs. Final price approximates the probability of an event occurring.

NFT PREDICTABILITY

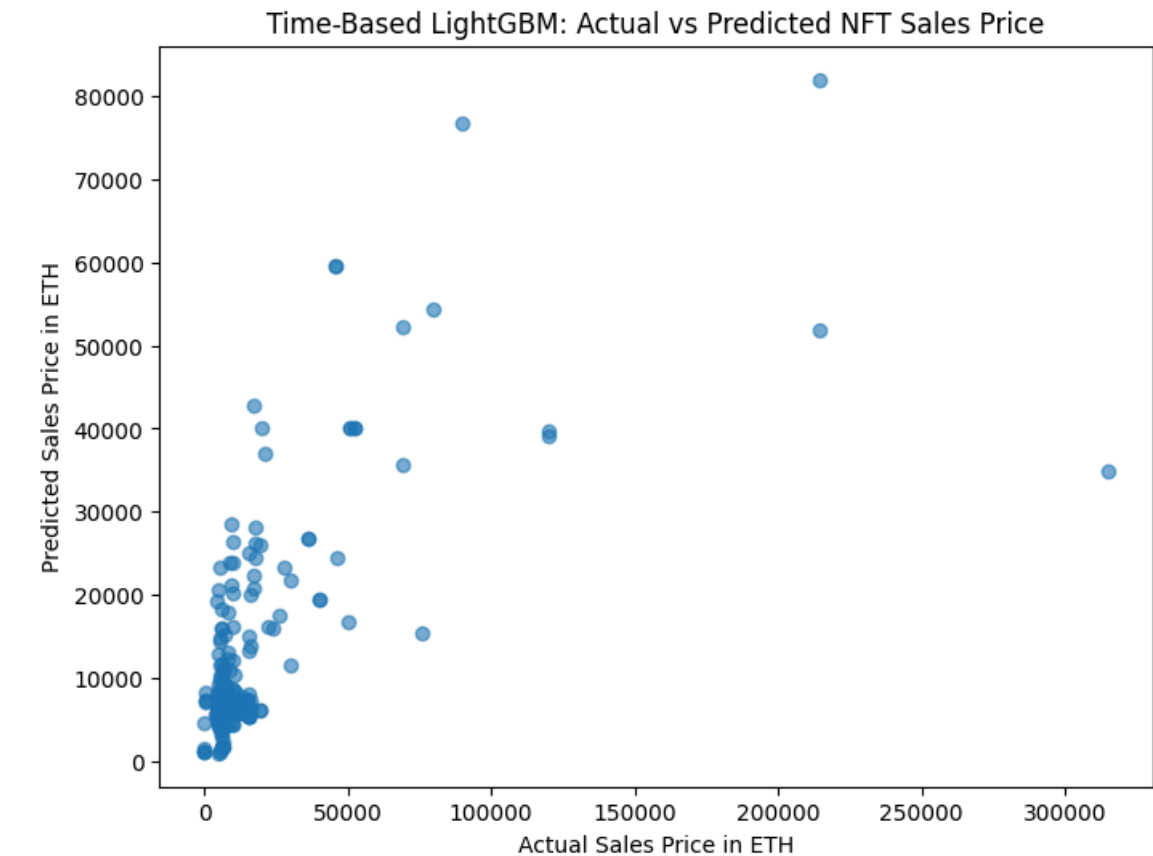
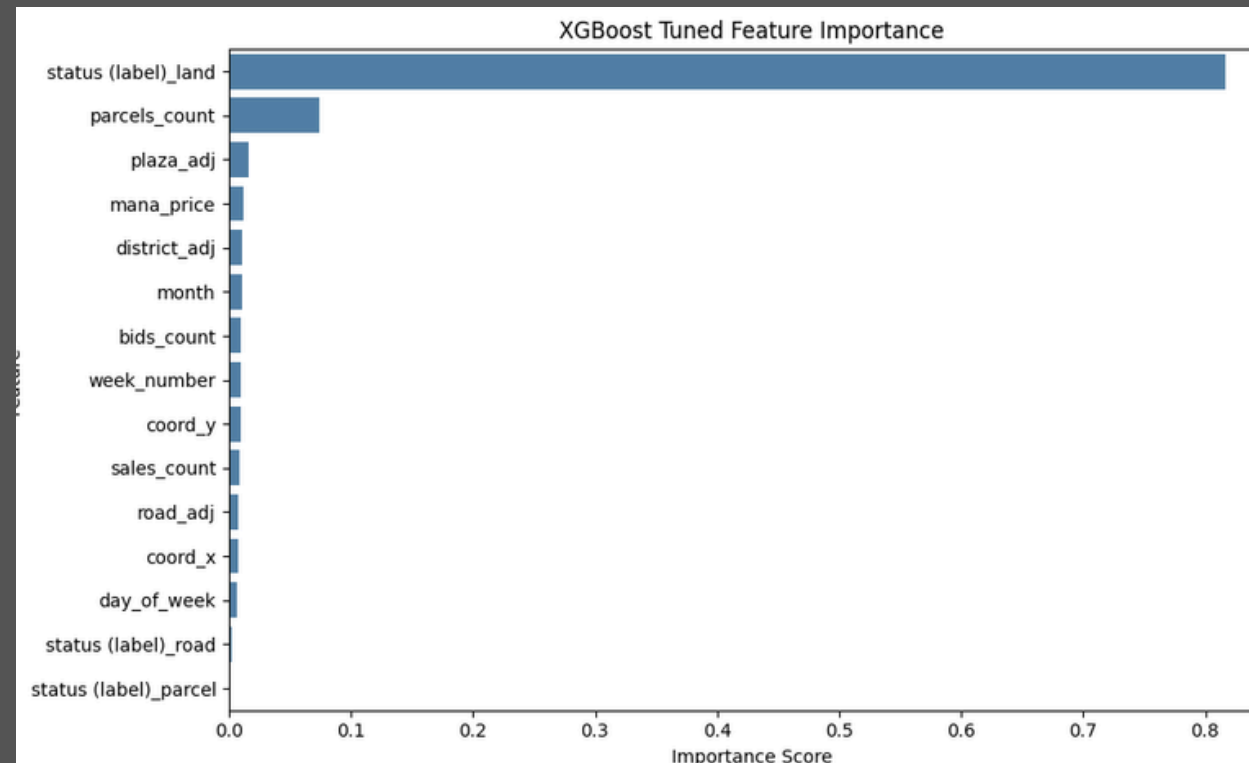
NFTs should be predictable because of efficient market hypothesis, prices reflect all available information, and financial incentives. There are also limits to predictability like bias, manipulation, and low participation.

OUR DATA SET:

Our data set consists of latest transactions (buy/sell) on the Decentraland's "land" datas, which have one of the highest bid/sell values in NFT and Metaverse projects. Similar to a real estate data, NFT land data consists of a house feature such as area, adjacency to important locations, Total number of buy/sells. Each "Land" consists of at least one parcel (the base metric value of a land in Decentraland), and can be interpreted as the Area of it.

OUR RESULTS

Status (label)_land dominates price prediction with an importance score of 0.816, confirming that estate classification is the primary value driver in the Decentraland market. Parcels_count follows at 0.074. This confirms that temporal market conditions influence NFT estate pricing — a finding unique to our combined AI-Finance modeling approach.



The chart above shows that the model performs better for lower-priced NFT sales, where most transactions are concentrated. However, it struggles with extreme high-value sales, which create large prediction errors.

KEY TAKEAWAYS

Prices approximate probabilities in theory. Markets perform best when participation is high, opinions are diverse, and Information is widely available.